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GEAR DATA  
NUMBER OF TEETH: 120  
DIAMETRAL PITCH: 38.00  
MODULE (mm, REF): 0.668  
PRESSURE ANGLE: 14.5 DEG  
PITCH DIAMETER (mm, REF): 80.21  
OUTSIDE DIAMETER (mm): 81.55 +0/-0.10  
WHOLE DEPTH (mm): 1.44 REF  
FACE WIDTH (mm): 3.00  
TOOTH FORM: INVOLUTE, FULL DEPTH

+0.05  
Ø 5.00 +0.03  
THRU - REAM

A

⊥ 0.05 A

Ra 1.6

CUT TEETH PER GEAR DATA.  
THIN DISC GEAR; GEAR TEETH: CIRCULAR RUNOUT 0.05 MAX ABOUT DATUM A, MEASURED AT THE TOOTH TIPS.  
DRIVEN 12:120 BY THE TRANSGEAR PINION (MHA-080).  
LOCKED COAXIAL TO THE 12T FEED PINION (MHA-110).

|                                                              |               |                            |                                                                                |                                                                        |                                 |               |            |                                                                                       |
|--------------------------------------------------------------|---------------|----------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------|---------------|------------|---------------------------------------------------------------------------------------|
| UNLESS OTHERWISE SPECIFIED, TOLERANCES                       |               |                            |                                                                                | Albert Michelson's Harmonic Analyzer<br>A Project For Hobby Machinists |                                 |               |            |                                                                                       |
| .XX<br>±0.51                                                 | .XXX<br>±0.13 | DRILLED HOLES<br>+0.10 / 0 | <div><div>≤</div><div>±1°</div></div> <div><div>√</div><div>Ra 3.2</div></div> |                                                                        |                                 |               |            |                                                                                       |
| REMOVE BURRS AND BREAK SHARP EDGES R0.25 OR CHAMFER 0.25 MAX |               |                            |                                                                                |                                                                        | PART<br><br>rack-pinion         |               |            |                                                                                       |
| FINISH<br>gear teeth cut; polished brass                     |               |                            |                                                                                |                                                                        |                                 |               |            |                                                                                       |
| MATERIAL<br>Brass                                            |               |                            |                                                                                |                                                                        | DWG. NO.<br><br>MHA-070v32      | REV<br><br>32 | SCALE: 1:1 |                                                                                       |
| INTERPRET GEOMETRIC TOLERANCING<br>PER: ASME Y14.5-2018      |               |                            |                                                                                |                                                                        |                                 |               | UNIT: mm   |                                                                                       |
| DO NOT SCALE DRAWING                                         |               |                            |                                                                                |                                                                        | © 2026 Pedro Paulo Vezza Campos |               |            |  |